



ENGO 545 – Fall 2026

Hydrographic Surveying

Laboratory exercise #1 (part 1)

Tide prediction



Deadline: Mon, Oct 5, 2026 (11:59 PM)

Learning objectives:

After completing this laboratory exercise students should be able to:

- Describe the mathematical model relating height of tide and harmonic constituents
- Look up harmonic constituents, tidal datums, and other information pertinent for tide prediction
- Classify the tidal regime, conduct tide prediction, and analyse tidal variations for a US tidal station
- Program an algorithm in MATLAB, Python, or C/C++
- Write a technical report

Background information:

The water level at a given tidal station can be predicted based on a number of previously estimated parameters. The following mathematical model can be used to compute water level for a time in the future from n harmonic constituents:

$$h(t) = H_0 + \sum_{i=1}^n \{f_i \cdot H_i \cdot \cos[a_i \cdot t + (V_0 + u)_i - \kappa_i]\}$$

where:

$h(t)$ is the height (above a reference datum) of the tide at time t [in metres or feet]

H_0 is the mean height of water level above the datum [in metres or feet]

f_i is the node factor for tidal constituent i

H_i is the amplitude for tidal constituent i [in metres or feet]

a_i is the angular speed of tidal constituent i (i.e., its frequency) [in degrees/hour]

t is the time, reckoned from some initial epoch (e.g., $t = 0$ at the beginning of the year) [in hours]

$(V_0 + u)_i$ is the equilibrium argument for tidal constituent i [in degrees]

κ_i is the epoch (phase lag) of tidal constituent i [in degrees] relative to the Moon's transit over the tide station

The mathematical model above is taken from the National Oceanic and Atmospheric Administration (NOAA) National Ocean Service (NOS) Center for Operational Oceanographic Products and Services (CO-OPS) Special Publication 3, which is available for download through the link below:

https://tidesandcurrents.noaa.gov/publications/Tidal_Analysis_and_Predictions.pdf

The value for H_0 at a particular tidal station is computed by subtracting the height of the desired datum from the mean water level (MWL) at that station. In the United States, the desired datum is the mean lower low water (MLLW), and the MWL is labeled as the mean sea level (MSL).

Values for f_i and $(V_0 + u)_i$ are generated from astronomical equations. They are published in Tables 14 and 15, respectively, of the NOAA NOS CO-OPS Special Publication No. 98 and its supplement:

<https://tidesandcurrents.noaa.gov/publications/SpecialPubNo98.pdf> (mid-1850s - 2000)
<https://tidesandcurrents.noaa.gov/publications/SpecialPubNo98Suppl.pdf> (2000 - 2025)

or in a data file* with a “compressed format”.

Values for the tidal harmonic constituents H_i , a_i , and κ_i are also needed. NOAA provides these values (termed amplitude, speed, and phase, respectively) for all United States tidal stations. For example, stations can be selected through the following link: https://tidesandcurrents.noaa.gov/tide_predictions.html. For a particular station, click on the “Harmonic Constituents” dropdown menu under the “Tides/Water Levels” tab.

Tasks:

Using the given mathematical model and lists of values, conduct the following tide prediction (in feet) for tidal station Seattle, WA (# 9447130) assuming January 1, 2025 (or 2026 for a bonus; see footnote *) as the initial epoch using a programming language of your choice (e.g., MATLAB, Python, or C/C++):

1. **Identify the appropriate tidal constituents and classify the tidal regime for the tidal station.**
2. Generate time series of the tidal variations for the first 72 hours in November 2026, the last 61 days (i.e., the months of November and December) of 2026, and the entire year 2026.
3. Compute the range of tide (based on the difference between the highest high water and the lowest low water) for each of those periods. Do you see a trend?
4. Explain the nature of the (semi-)daily, fortnightly or monthly, and/or any other variations that you notice from the plots. **Back up your analysis with the tidal regime classification you performed in 1)** as well as the magnitude of any other dominant tidal harmonic constituents.

Deliverables:

Check the Additional Course Information document on D2L for the required deliverables on a laboratory exercise.

* This is an ASCII file with 80 characters per line, no separators between the different values on a line; 16 lines per year, providing the values for up to 128 constituents; the order of the constituents is not known; here is a best guess in terms of the order of the characters in each line:

- Columns 1-4 == year
- Column 5 == leap year indicator (1 or 0)
- Columns 6-8 == line number (01 to 16)
- Columns 9-12 == Table 14 value for a constituent (given to 3 decimal places)
- Columns 13-16 == Table 15 value for a constituent (given to 1 decimal place)
- Columns 17-72 == repeat of columns 9-16 for the rest of the constituents in an 8-character format
- Columns 73-80 == end of line indicators

A bonus will be awarded if the Table 14 and Table 15 values for 2026 are correctly extracted and used for the prediction